Jiachi Zhang

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EDUCATION

South China University of Technology **09/2023 - 06/2027**

Qualification: Bachelor of Engineering

Major: Information Engineering

Average Score: 84.52/100

Core Courses: Signals & Systems, Analog Electronics, Digital Electronics, Digital Signal Processing, Principle of Communications, Microcomputer System and Interface Technology, Programming in C++

SUMMER STUDY PROGRAM

École Polytechnique de l'Université de Nantes, France **07/2025**

Core Courses: Français, Théorie de l'Information

INTERNSHIP EXPERIENCE

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences **10/2025 - Present**

Position: Visiting Research Intern (Advisors: Dr. Wenqi Fang & PhD candidate Zhuoyu Wu)

- Carried out independent research by identifying research gaps, developing research hypotheses, designing experimental protocols, and conducting hands-on experiments and data analysis.
- Compiled academic materials by reviewing relevant literature and drafting concise and structured experimental summaries.
- Authored and revised the thesis, ensuring a clear, logically rigorous scientific narrative grounded in experimental results.

PUBLICATIONS & RESEARCH

Gain-Aware Prediction-Space Recursive Controller for Lightweight Polyp Segmentation **09/2025 - Present**

Submitted to ICONIP 2026.

- Formulated segmentation refinement as a prediction-space recursive correction task, shifting correction entirely into logit space to avoid re-engaging heavy feature hierarchies.
- Designed a three-stage recursive controller (Prediction Evidence Encoding → Gain-Aware State Update → State-Guided Residual Correction) with only 189 parameters and negligible GMACs overhead.
- Evaluated across 7 lightweight backbones (UltraLBM-UNet, Mobile-PolypNet, EGE-UNet, CGNet, ULite, CMUNeXt-S, I2U-Net) on 4 polyp segmentation datasets under a unified Kvasir-trained protocol.
- Achieved consistent source-domain improvements and competitive performance against both training-side baselines (+LL/+DS/+Mutation/+LoMix) and the heavier

structural refiner +Harmonizing.

RIGS-Refiner: Risk-Guided Recursive Refinement in Prediction Space for Polyp Segmentation **10/2025 - Present**

Submitted to APSIPA ASC 2026.

- Proposed a lightweight post-refinement plugin for risk-guided recursive correction directly in prediction space, built on frozen host predictions.
- Designed a shared recursive cell combining image priors (Sobel edge, average pooling), prediction cues, risk-guided update, and residual correction with only 519 parameters and +0.631 GFLOPs.
- Tested on PraNet and SegFormer-B0 across Kvasir-SEG, ClinicDB, ColonDB, and ETIS, achieving consistent IoU/HD95/clDice improvements with a favorable efficiency–accuracy trade-off against SegFix, rNCA, and CascadePSP.

Glomerular Pathological Image Detection via YOLO11 with Stain Normalization **04/2024 - 03/2025**

- Designed color clustering and color transfer modules to mitigate staining variance and augment glomerular pathological image datasets.
- Constructed a YOLO11-based detection framework and adopted Focal Loss to balance positive-and-negative samples in the glomerular detection task.
- Integrated EMA multi-scale attention to enhance feature fusion and suppress tissue noise; validated via ablation experiments, achieving 94% mAP that outperformed baseline models.

PROJECT EXPERIENCE

Macroeconomic Indicator Forecasting System via SAITS and Time-Series LLMs **07/2025 - 08/2025**

- Adopted the SAITS algorithm to conduct multivariate time-series imputation, and reconstructed high-fidelity continuous feature spaces under limited labeled data.
- Integrated TimeLLM and ChatTS, designed cross-modal alignment modules to capture temporal evolution trends and enhance prediction accuracy.
- Constructed a modular training and evaluation system, and developed high-precision interpretable time-series models, securing top 10 in the E Fund FinTech Challenge.

FPGA-Based Shooting Confrontation Game **03/2026**

- Developed a dual-mode FPGA shooting game supporting 1280×800@60fps HDMI output, infrared control and real-time score display.
- Designed HDMI display drivers and SDRAM caching logic, and implemented AABB-based collision detection for game objects.
- Used finite state machines to control seven game states and ensure synchronization between peripherals and display modules.
- Completed project validation and received the Excellent Course Design award for outstanding academic performance.

EXTRACURRICULAR EXPERIENCE

President, Art Troupe, Youth League Committee **11/2024 - 11/2025**

- Participated in the production and singing of the department song, and performed in the college band on various occasions.

- Managed and coordinated internal departmental affairs, and took charge of the college graduation gala as a key organizer for multiple times.

Study Representative, Class Committee **09/2023 - 10/2025**

- Collected and distributed daily assignments, and completed the liaison work for academic affairs.
- Communicated actively between teachers and classmates, and released official university notifications in a timely manner.

HONORS & AWARDS

3rd Prize Scholarship, South China University of Technology **12/2025**

Merit Student, South China University of Technology **12/2025**

Advanced Individual in Student Work, South China University of Technology **09/2025**

Active Participant in Student Work, South China University of Technology **09/2025**

10th Place, Pazhou Algorithm Competition - E Fund Cup FinTech Challenge **08/2025**

Outstanding League Cadre, South China University of Technology **05/2025**

ADDITIONAL SKILLS

Language Ability: French (basic), English (fluent, can be used as a studying language)

Programming & Development: Python, PyTorch, Keil, MATLAB, Vivado